Test Cases

For

Custom Shell and Utilities

CSC 332- Operating Systems I

4/23/26

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Myshell Interface Testing:

Test #	Input	Expected Output	Actual Output	Pass/Fail
S-01	help	Display list of 5 utilities & syntax.	Help menu displayed with flags.	Pass
S-02	ls	Error: 'ls' is not whitelisted. Type 'help' for list.	Error message displayed.	Pass
S-03	history	List of previous commands.	Numbered list of 50 commands.	Pass
S-04	(loganalyzer -f logtiny.log)	Run command (strip parentheses).	Success; parentheses ignored.	Pass
S-05	sleep 10 &	Prompt returns immediately (PID shown).	[Process running in background]	Pass

Screenshots successfully running:

```
myshell> help


--- Custom Shell Command Guide ---
Format: <command> [flags] [arguments]

1. loganalyzer      : loganalyzer -f <filename>
2. filecrypt        : filecrypt -f <filename> -k <key>
3. timedexec        : timedexec -t <wall seconds> -c <CPU seconds> -m <MB memory> -- <command>
4. filediffadvanced : filediffadvanced -a <file1> -b <file2>
5. processgroup     : processgroup -g <pgid> -s <signal> -w <watch group seconds>

Built-in: help, exit
---------------------------------
```

```
myshell> history

1: ls
2: help
3: help
4: loganalyzer -f loganalyzertest.log
5: history
myshell>
```

```
myshell> asdasd

Error: 'asdasd' is not whitelisted. Type 'help' for list.
myshell>
```

```
myshell> sleep 10 &sleep 10 &

[Process running in background with PID 896]
[Process running in background with PID 897]
```

Loganalyzer Testing:

Test #	Input	Expected Output	Actual Output	Pass/Fail
L-01	loganalyzer -f logtiny.log	Analysis Complete: 42 bytes, 6 lines found. Execution Time:	Analysis Complete: 42 bytes, 6 lines found. Execution Time: 0.000621 seconds	Pass
L-02	loganalyzer -f loglarge.log	Analysis Complete: 112671 bytes, 4050 lines found. Execution Time:	Analysis Complete: 112671 bytes, 4050 lines found. Execution Time: 0.000804 seconds	Pass
L-03	loganalyzer -f ghost.log	open: No such file or directory	open: No such file or directory	Pass
L-04	loganalyzer -g test.log	Usage: loganalyzer -f <filename> myshell> loganalyzer: unknown option -- g	Usage: loganalyzer -f <filename> myshell> loganalyzer: unknown option -- g	Pass
L-05	loganalyzer -f /cygdrive/c/Users/dakot/Desktop/testing1.log	Analysis Complete: 41 bytes, 7 lines found. Execution Time:	Analysis Complete: 41 bytes, 7 lines found. Execution Time: 0.000700 seconds	Pass

Screenshots successfully running:

```
myshell> loganalyzer -f logtiny.log

Analysis Complete: 42 bytes, 6 lines found.
Execution Time: 0.000621 seconds
myshell> loganalyzer -f loglarge.log

Analysis Complete: 112671 bytes, 4050 lines found.
Execution Time: 0.000804 seconds
myshell>
```

```
myshell> loganalyzer -f /cygdrive/c/Users/dakot/Desktop/testing1.log

Analysis Complete: 41 bytes, 7 lines found.
Execution Time: 0.000700 seconds
myshell>
```

```
myshell> loganalyzer -g test..log

Usage: loganalyzer -f <filename>
myshell> loganalyzer: unknown option -- g
```

```
myshell> loganalyzer -f ghost.log

myshell> open: No such file or directory
```

Filecrypt Testing

Test 1

Input:

filecrypt -f regulartext.txt -k CSC332 > encryptedtext.bin

filecrypt -f encryptedtext.bin -k CSC332

Expected Output: Hello, World!

Actual Output:

```
myshell> filecrypt -f regulartext.txt -k CSC332 > encryptedtext.bin

6/_\c,A_VbYmyshell> filecrypt -f encryptedtext.bin -k CSC332
Hello, World!
myshell>
```

Pass

Test 2

Input:

filecrypt -f regulartext.txt -k Pickle > encryptedtext2.bin

filecrypt -f encryptedtext2.bin -k Pickle

Expected Output: Hello, World!

Actual Output:

```
   qcmyshell> filecrypt -f regulartext.txt -k Pickle > encryptedtext2.bin

Ip>
   qcmyshell> filecrypt -f encryptedtext2.bin -k Pickle
Hello, World!
myshell>
```

Pass

Test 3

Input:

filecrypt -f regulartext.txt -k secret> encryptedtext3.bin

filecrypt -f encryptedtext3.bin -k secret

Expected Output: Hello, World!

Actual Output:

```
myshell> filecrypt -f regulartext.txt -k secret > encryptedtext3.bin
;
XS2
        Romyshell> filecrypt -f encryptedtext3.bin -k secret
Hello, World!
imyshell> |
```

Timedexec Testing

========== timedexec — Test Log

==========

HOW TO COMPILE

Mac/Linux: gcc -Wall -O2 -o timedexec timedexec.c -lpthread

```
● tav@MacBookPro CSC332Midterm % gcc -Wall -O2 -o timedexec timedexec.c -lpthread
```

Windows:

gcc -Wall -O2 -o timedexec timedexec.c -lpthread (run inside WSL terminal, same command)

==========

TEST 1: Wall-clock limit

Mac/Linux: ./timedexec -t 3 -- sleep 10

Windows: timedexec.exe -t 3 -- timeout /t 10

--- Performance Metrics ---

Wall Time : 2.521 s

CPU (user) : 0.001 s

CPU (sys) : 0.002 s

Peak RSS : 1032192 kB

Killed by : wall-clock limit

Exit code: 124

RESULT: PASS

```
tav@MacBookPro CSC332Midterm % ./timedexec -t 3 -- sleep 10

--- Performance Metrics ---
Wall Time    : 2.521 s
CPU (user)   : 0.001 s
CPU (sys)    : 0.002 s
Peak RSS     : 1032192 kB
Killed by    : wall-clock limit
```

---------- TEST 2: Normal exit, no limits hit

Mac/Linux: ./timedexec -t 10 -- sleep 2

Windows: timedexec.exe -t 10 -- timeout /t 2

--- Performance Metrics ---

Wall Time : 2.521 s

CPU (user) : 0.001 s

CPU (sys) : 0.002 s

Peak RSS : 917504 kB

Exit code: 0

RESULT: PASS

```
tav@MacBookPro CSC332Midterm % ./timedexec -t 10 -- sleep 2

--- Performance Metrics ---
Wall Time    : 2.521 s
CPU (user)   : 0.001 s
CPU (sys)    : 0.002 s
Peak RSS     : 1032192 kB
```

TEST 3: No command given (error handling)

Mac/Linux: ./timedexec -t 5

Windows: timedexec.exe -t 5

Error: No command specified.

Exit code: 1

RESULT: PASS

```
⊗ tav@MacBookPro CSC332Midterm % ./timedexec -t 5
  Error: No command specified.
```

TEST 4: Bad command (error handling)

Mac/Linux: ./timedexec -t 5 -- fakecommand

Windows: timedexec.exe -t 5 -- fakecommand

---

execvp: No such file or directory

Exit code: 127

RESULT: PASS

```
tav@MacBookPro CSC332Midterm % ./timedexec -t 5 -- fakecommand
execvp: No such file or directory

--- Performance Metrics ---
Wall Time   : 0.512 s
CPU (user)  : 0.000 s
CPU (sys)   : 0.002 s
Peak RSS    : 868352 kB
```

==========

Results: 4 passed / 0 failed / 4 total

==========

FileDiffAdvanced Testing

fileDiffAdvanced Test Cases:

Successful txt w/ txt:

Input:

$ ./filediffadvanced -f test1.txt -k test2.txt

Output:

--- Performance Metrics ---

Total Edit Distance: 30

Execution Time: 0.000296 seconds

Program ended with exit code: 0

Unsuccessful txt w/ other:

Input:

$ ./filediffadvanced -f test1.txt -k test2.mp4

Output:

ERROR!!

Compared files have to be same.

Program ended with exit code: 255

Unsuccessful other w/ other:

Input:

$ ./filediffadvanced -f test1.mp4 -k test2.mp4

Output:

ERROR!!

Only .txt files allowed.

Program ended with exit code: 255

Processgroup Testing

TEST #	*Input*	*Expected Output*	*Actual Output*	*Pass/Fail*
1	./processgroup	Error message about missing -g	Error: -g <pgid> is required.	PASS
2	./processgroup -g 234082	Lists bash process with memory	PID 234082 bash 5668kB	PASS
3	./processgroup -g 99999	No processes found	0 process(es) found	PASS
4	./processgroup -g 234082 -w 4	4 monitor lines then done	[monitor] 4 lines printed	PASS
5	./processgroup -g 234082 -s 15	Signal sent successfully	Signal 15 sent successfully	PASS
6	./processgroup -z	Invalid option error	invalid option -- 'z'	PASS

#1:

```
[ouhabaz@www ~]$ ./processgroup
Error: -g <pgid> is required.
Usage: ./processgroup -g <pgid> [-s <signal>] [-w <seconds>]
```

#2:

```
[ouhabaz@www ~]$ ./processgroup -g 234082

Processes in group 234082:
  PID         Name                        Memory
  ------------------------------------------------------
  PID 234082  Name: bash                    Memory (RSS): 5668 kB

Total: 1 process(es) in group 234082
```

#3:

```
[ouhabaz@www ~]$ ./processgroup -g 99999

Processes in group 99999:
  PID         Name                        Memory
  ----------------------------------------------------
  (no processes found in group 99999)

Total: 0 process(es) in group 99999
```

#4:

```
[ouhabaz@www ~]$ ./processgroup -g 234082 -w 4

Processes in group 234082:
  PID         Name                        Memory
  ----------------------------------------------------
  PID 234082  Name: bash                  Memory (RSS): 5668 kB

Total: 1 process(es) in group 234082

Watching group 234082 for 4 second(s)...

[monitor] Group 234082 -> 1 process(es) alive
[monitor] Group 234082 -> 1 process(es) alive
[monitor] Group 234082 -> 1 process(es) alive
[monitor] Group 234082 -> 1 process(es) alive

Watch complete. Last recorded count: 1 process(es).
[1]+  Done                    sleep 999
```

#5:

```
[ouhabaz@www ~]$ ./processgroup -g 234082 -s 15

Processes in group 234082:
  PID         Name                        Memory
  ----------------------------------------------------
  PID 234082  Name: bash                  Memory (RSS): 5668 kB

Total: 1 process(es) in group 234082

Sending signal 15 to process group 234082...
Signal 15 sent successfully.
```

#6:

```
[ouhabaz@www ~]$ ./processgroup -z
./processgroup: invalid option -- 'z'
Usage: ./processgroup -g <pgid> [-s <signal>] [-w <seconds>]
  -g  Process group ID (required)
  -s  Signal number to send (e.g. 15 = SIGTERM)
  -w  Watch group live for N seconds (1 update/sec)
```